

Una Aplicación del Análisis Complejo al Álgebra Lineal

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I Semana de la Ciencia
Universidad Distrital Francisco José de Caldas
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I PARTE: TEOREMA DE CAUCHY



Limonos Amarillos. Carlos Rojas González.



Cauchy

THÉORÈME 1.^{er}

Soit $f(x) = p$ une fonction de x telle que, si l'on fait

$$f(x \pm \varepsilon \sqrt{-1}) = P' \pm P'' \sqrt{-1},$$

P' et P'' conservent une valeur déterminée pour toutes les valeurs de x et de ε comprises entre les limites $x = 0$, $x = a$, $\varepsilon = 0$, $\varepsilon = b$; et qu'en outre P'' s'évanouisse aux deux limites de x , quelle que soit d'ailleurs la valeur de ε . Si l'on suppose dans P' , $\varepsilon = b$, on aura

$$\int P' dx = \int P dx,$$

les intégrales étant prises entre les limites $x = 0$, $x = a$.

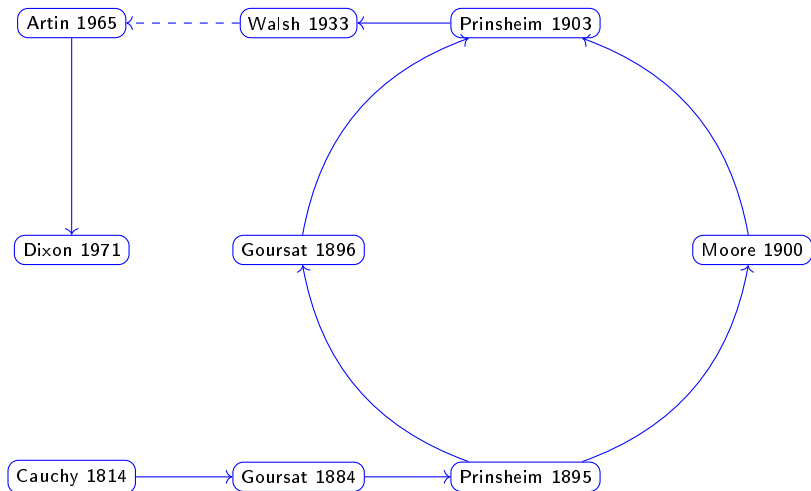
*Memoria sobre integrales definidas.
Presentado 1814, publicado 1825*

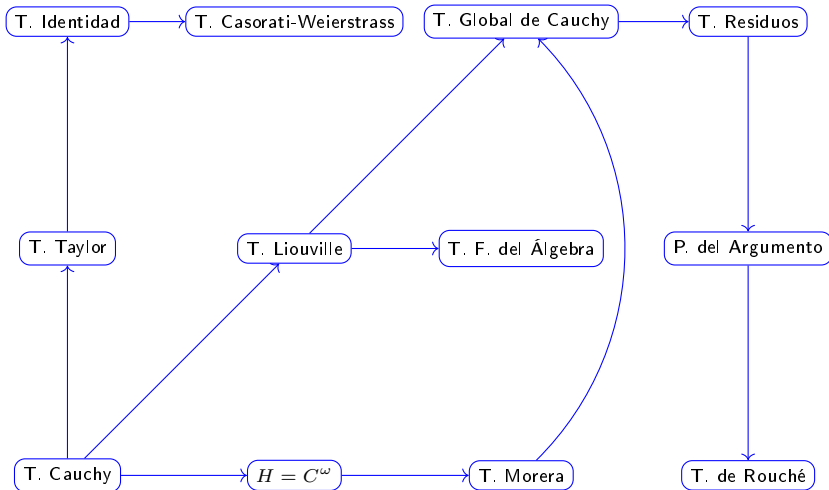


Poisson



Legendre



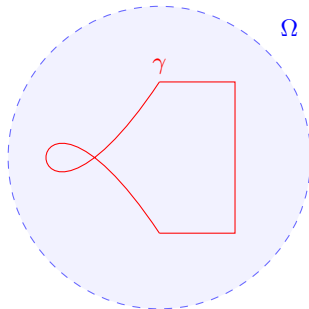


Teorema (Cauchy)

Sea $\Omega \subset \mathbb{C}$ un bola abierta del plano complejo. Si $f \in H(\Omega)$, entonces

$$\int_{\gamma} f(z) dz = 0$$

para todo arco γ cerrado y diferenciable a trozos.



Porisma

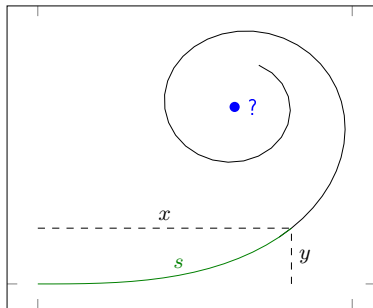
Encontrar el destino de la trayectoria de una carretera curva que hace constante la sobreaceleración (sin sacudones) al interior de los vehículos que la transitan



Puente Risaralda-Manizalez. Adobe Stock.

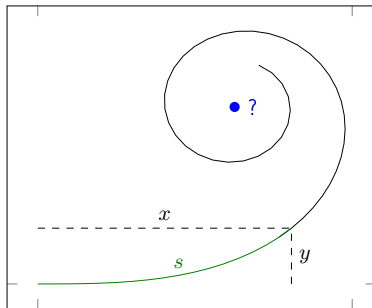
$$a'(s) = k'(s) = 2$$

Destino de una carretera sin sacudones



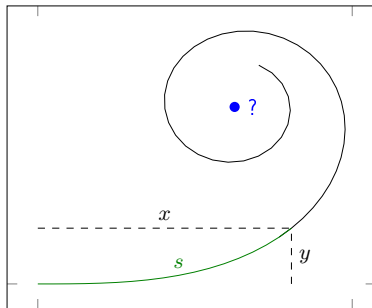
$$a(s) = \sqrt{x''(s)^2 + y''(s)^2} = 2s$$

Destino de una carretera sin sacudones



$$x''(s)^2 + y''(s)^2 = 4s^2$$

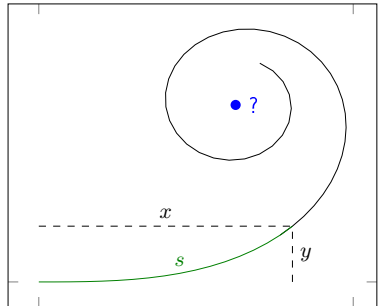
Destino de una carretera sin sacudones



$$x''(s)^2 + y''(s)^2 = 4s^2$$

$$x'(s)^2 + y'(s)^2 = 1$$

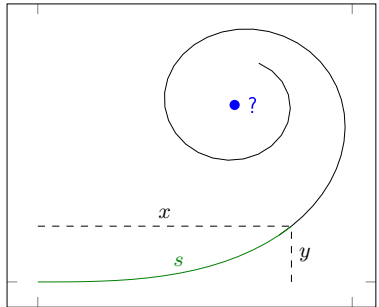
Destino de una carretera sin sacudones



$$x''(s)^2 + y''(s)^2 = 4s^2$$

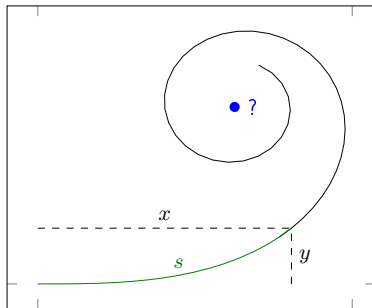
$$\begin{cases} x'(s) = \cos(\theta(s)) \\ y'(s) = \sin(\theta(s)) \end{cases}$$

Destino de una carretera sin sacudones



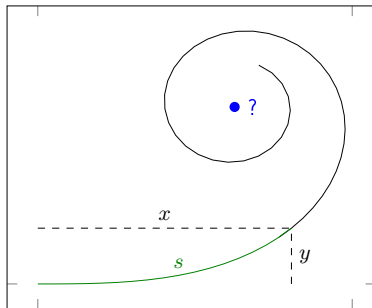
$$\begin{aligned}\theta'(s) &= 2s \\ \begin{cases} x'(s) = \cos(\theta(s)) \\ y'(s) = \sin(\theta(s)) \end{cases}\end{aligned}$$

Destino de una carretera sin sacudones



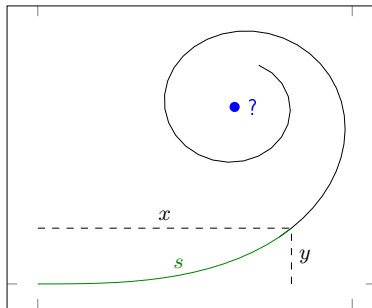
$$\begin{cases} x'(s) = \cos(s^2) \\ y'(s) = \sin(s^2) \end{cases}$$

Destino de una carretera sin sacudones



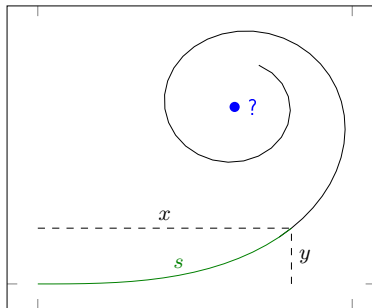
$$\begin{cases} x(s) = \int_0^s \cos(t^2) dt \\ y(s) = \int_0^s \sin(t^2) dt \end{cases}$$

Destino de una carretera sin sacudones



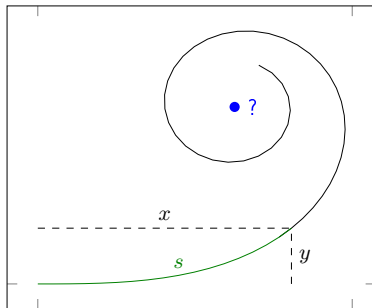
$$\int_0^\infty \cos(t^2) dt \quad \int_0^\infty \sin(t^2) dt$$

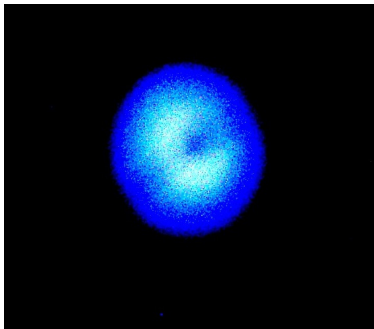
Destino de una carretera sin sacudones



$$\int_0^\infty \cos(t^2) dt = ? \quad \int_0^\infty \sin(t^2) dt = ?$$

Destino de una carretera sin sacudones





Difracción de Fresnel. Patrick Moran

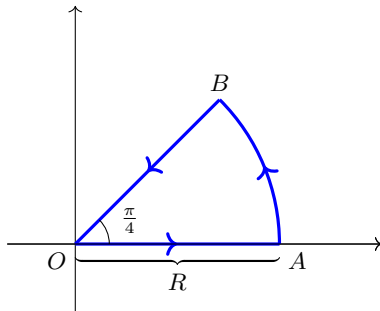


Salitre Mágico. Miguel Perez.

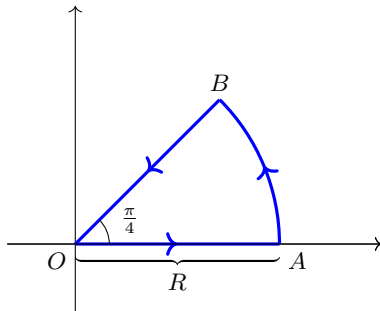
Consideremos

$$f(z) = e^{iz^2}$$

en $\mathbb{C}$ donde γ es la frontera del sector circular OAB .

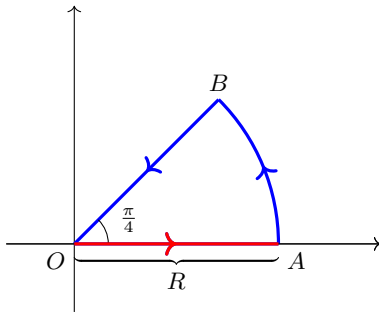


$$\begin{aligned}
 & \int_{\gamma} e^{iz^2} dz \\
 &= \int_{OA} e^{iz^2} dz + \int_{AB} e^{iz^2} dz + \int_{BO} e^{iz^2} dz \\
 &= 0
 \end{aligned}$$



$$z(t) = t, \quad t \in [0, R]$$

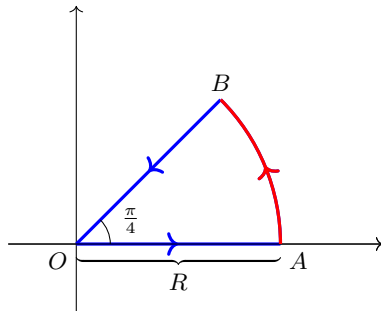
$$\begin{aligned} & \int_{OA} e^{iz^2} dz \\ &= \int_0^R e^{it^2} dt \\ &= \int_0^R \cos(t^2) dt + i \int_0^R \sin(t^2) dt \\ &\rightarrow \int_0^\infty \cos(t^2) dt + i \int_0^\infty \sin(t^2) dt \\ &\text{cuando } R \rightarrow \infty \end{aligned}$$



$$z(t) = Re^{it}, \quad t \in \left[0, \frac{\pi}{4}\right]$$

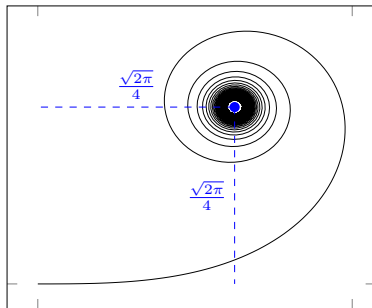
$$\int_{AO} e^{iz^2} dz = iR \int_0^{\frac{\pi}{4}} \exp(iR^2 e^{2it}) e^{it} dt$$

$\rightarrow 0$ cuando $R \rightarrow \infty$



$$\int_0^{\infty} \cos(t^2) dt = \int_0^{\infty} \sin(t^2) dt = \frac{\sqrt{2\pi}}{4}$$

Destino de una carretera sin sacudones



Teorema (Formula integral de Cauchy de orden superior)

Sea Ω un abierto conexo, $f \in H(\Omega)$ y $\overline{B_r(a)} \subset \Omega$. Si $z \in \Omega$, entonces

$$f^{(n)}(a) = \frac{n!}{2\pi i} \oint_{|z-a|=r} \frac{f(z)}{(z-a)^{n+1}} dz.$$

Ejemplo

Sea $\chi(z) = \sum_{j=0}^N a_j z^j$ y $R > 0$. Calculemos el valor de

$$S = \oint_{|z|=R} \frac{\chi(z)}{z^{n+1}} dz$$

$$\begin{aligned} \oint_{|z|=R} \frac{\chi(z)}{z^{n+1}} dz &= \oint_{|z|=R} \sum_{j=0}^N \frac{a_j z^j}{z^{n+1}} dz \\ &= \sum_{j=0}^N a_j \oint_{|z|=R} \frac{z^j}{z^{n+1}} dz \end{aligned}$$

Valor de S depende de

$$\oint_{|z|=R} \frac{z^j}{z^{n+1}} dz$$

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Valor de S depende de

$$\oint_{|z|=R} \frac{z^j}{z^{n+1}} dz$$

$$j > n$$

$$\oint_{|z|=R} \frac{z^j}{z^{n+1}} dz = 0$$

$$j = n$$

$$\oint_{|z|=R} \frac{1}{z} dz = \frac{2\pi i}{1!} \cdot 1$$

$$j < n$$

$$\oint_{|z|=R} \frac{z^j}{z^{n+1}} = \frac{2\pi i}{n!} \frac{d^n z^j}{dz^n} \Big|_{z=0}$$

$$j > n$$

$$\oint_{|z|=R} \frac{z^j}{z^{n+1}} dz = 0$$

$$j = n$$

$$\oint_{|z|=R} \frac{1}{z} dz = 2\pi i$$

$$j < n$$

$$\oint_{|z|=R} \frac{z^j}{z^{n+1}} = \frac{2\pi i}{n!} \left. \frac{d^n z^j}{dz^n} \right|_{z=0}$$

$$j > n$$

$$\oint_{|z|=R} \frac{z^j}{z^{n+1}} dz = 0$$

$$j = n$$

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$$j > n$$

$$\oint_{|z|=R} \frac{z^j}{z^{n+1}} dz = 0$$

$$j = n$$

$$\oint_{|z|=R} \frac{1}{z} dz = 2\pi i$$

$$j < n$$

$$\oint_{|z|=R} \frac{z^j}{z^{n+1}} = 0$$

Haciendo

$$\oint_{|z|=R} \frac{z^j}{z^{n+1}} dz = 2\pi i \delta_{jn}$$

Valor de la integral S

$$\sum_{j=0}^N a_j \oint_{|z|=R} \frac{z^j}{z^{n+1}} dz = \sum_{j=0}^N 2\pi i a_j \delta_{jn}$$

Haciendo

$$\oint_{|z|=R} \frac{z^j}{z^{n+1}} dz = 2\pi i \delta_{jn}$$

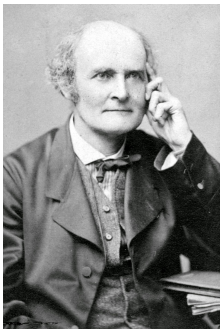
Valor de la integral S

$$\sum_{j=0}^N a_j \oint_{|z|=R} \frac{z^j}{z^{n+1}} dz = \sum_{j=0}^N 2\pi i a_j \delta_{jn}$$

II PARTE: TEOREMA DE CAYLEY-HAMILTON



Paisaje. Carlos Rojas González.



Cayley

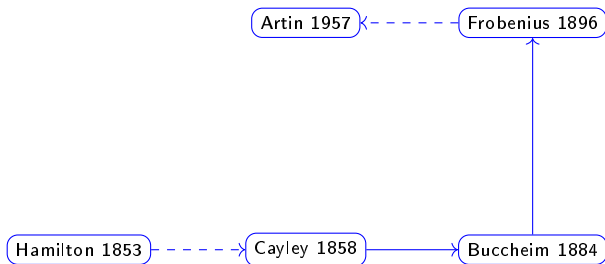
II. *A Memoir on the Theory of Matrices.* By ARTHUR CAYLEY, Esq., F.R.S.

Received December 10, 1857,—Read January 14, 1858.

THE term matrix might be used in a more general sense, but in the present memoir I consider only square and rectangular matrices, and the term matrix used without qualification is to be understood as meaning a square matrix; in this restricted sense, a set of quantities arranged in the form of a square, *e. g.*

$$\begin{pmatrix} a & b & c \\ a' & b' & c' \\ a'' & b'' & c'' \end{pmatrix}$$

*Una memoria sobre la teoría de matrices
1857, publicado 1858*



Theorem (Cayley-Hamilton)

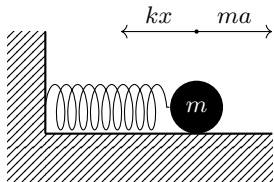
Sea A una matriz cuadrada con elementos en un cuerpo $\mathbb{K} = (\mathbb{C}, \mathbb{R})$ y $\chi(\lambda)$ el polinomio característico de A . Entonces, $\chi(A) = 0$. Es decir, toda matriz cuadrada es raíz de su polinomio característico.

Oscilador Armónico

$$F = -kx \quad F = ma = mx''(t)$$

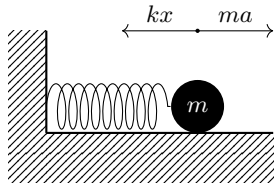
Frecuencia de oscilación

Hacemos $\omega := \sqrt{\frac{k}{a}}$



Oscilador Armónico

$$x'' + \omega^2 x = 0$$

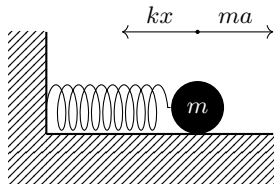


Oscilador Armónico

$$x'' + \omega^2 x = 0$$

Condiciones iniciales

$$x(0) = x_0 \text{ y } x'(0) = y_0.$$



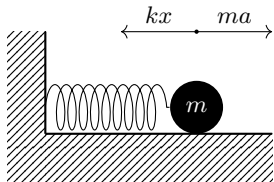
Oscilador Armónico

$$x'' + \omega^2 x = 0$$

Haciendo

$$x' = y$$

$$y' = x'' = -\omega^2 x$$

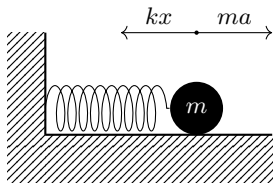


Obtenemos

$$\begin{pmatrix} x \\ y \end{pmatrix}' = \begin{pmatrix} 0 & 1 \\ -\omega^2 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

con condiciones iniciales

$$\begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$$



Obtenemos

$$\begin{pmatrix} x \\ y \end{pmatrix}' = \begin{pmatrix} 0 & 1 \\ -\omega^2 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

con condiciones iniciales

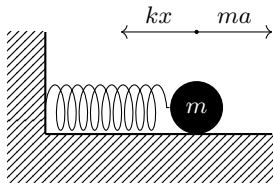
$$\begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$$

Haciendo

$$X = (x, y)$$

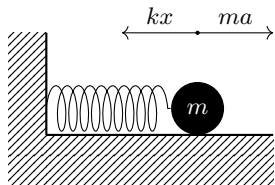
$$X_0 = (x_0, y_0)$$

$$A = \begin{pmatrix} 0 & 1 \\ -\omega^2 & 0 \end{pmatrix}$$



Problema de valor inies Iniciales

$$\begin{cases} X'(t) = AX(t) \\ X(0) = X_0 \end{cases}$$

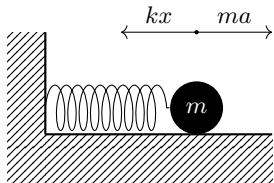


Problema de valor inies Iniciales

$$\begin{cases} X'(t) = AX(t) \\ X(0) = X_0 \end{cases}$$

Solución

$$X(t) = e^{At} X_0$$

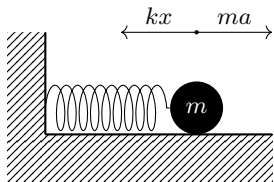


Problema de valor inies Iniciales

$$\begin{cases} X'(t) = AX(t) \\ X(0) = X_0 \end{cases}$$

Solución

$$X(t) = e^{At} X_0 = \left(\sum_{n=0}^{\infty} \frac{t^n A^n}{n!} \right) X_0$$

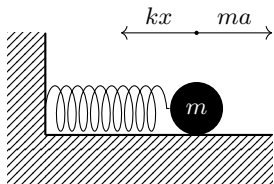


Obtenemos

$$A = \begin{pmatrix} 0 & 1 \\ -\omega^2 & 0 \end{pmatrix}$$

Polinomio Característico

$$\chi(\lambda) = \det(A - \lambda I)$$

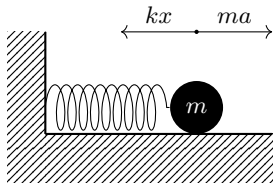


Obtenemos

$$A - \lambda I = \begin{pmatrix} -\lambda & 1 \\ -\omega^2 & -\lambda \end{pmatrix}$$

Polinomio Característico de A

$$\chi(\lambda) = \omega^2 + \lambda^2$$



Obtenemos

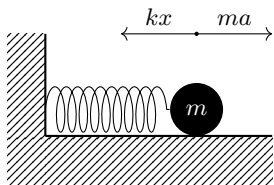
$$A - \lambda I = \begin{pmatrix} -\lambda & 1 \\ -\omega^2 & -\lambda \end{pmatrix}$$

Polinomio Característico de A

$$\chi(\lambda) = \omega^2 + \lambda^2$$

Por el Teorema de Cayley-Hamilton

$$A^2 = -\omega^2 I$$



Potencias de A

$$A^2 = -\omega^2 I \quad A^3 = -\omega^2 A$$

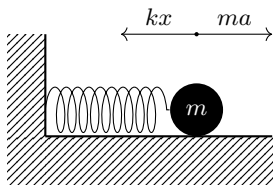
$$A^4 = \omega^4 I \quad A^5 = \omega^4 A$$

$$A^6 = -\omega^6 I \quad A^7 = -\omega^6 A$$

Sucesión de potencias de A

$$A^{2n} = (-1)^n \omega^{2n} I$$

$$A^{2n+1} = (-1)^n \omega^{2n} A$$



$$A^{2n} = (-1)^n \omega^{2n} I \quad A^{2n+1} = (-1)^n \omega^{2n} A$$

Serie de potencias de e^{At}

$$\begin{aligned} e^{At} &= \sum_{n=0}^{\infty} \frac{t^n A^n}{n!} \\ &= \sum_{n=0}^{\infty} \frac{t^{2n} A^{2n}}{(2n)!} + \sum_{n=0}^{\infty} \frac{t^{2n+1} A^{2n+1}}{(2n+1)!} \\ &= \sum_{n=0}^{\infty} I \cdot \frac{(-1)^n (\omega t)^{2n}}{(2n)!} + \frac{1}{\omega} \sum_{n=0}^{\infty} A \cdot \frac{(-1)^n (\omega t)^{2n+1}}{(2n+1)!} \\ &= I \cos(\omega t) + A \omega^{-1} \sin(\omega t) \end{aligned}$$

$$A^{2n} = (-1)^n \omega^{2n} I \quad A^{2n+1} = (-1)^n \omega^{2n} A$$

Serie de potencias de e^{At}

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Serie de potencias de e^{At}

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Solución

$$X(t) = e^{At} X_0$$

Valor inicial

$$X_0 = (x_0 \quad y_0)^t$$

$$e^{At} = I \cos(\omega t) + A\omega^{-1} \sin(\omega t)$$

$$= \begin{pmatrix} \cos(\omega t) & 0 \\ 0 & \cos(\omega t) \end{pmatrix} + \begin{pmatrix} 0 & \omega^{-1} \sin(\omega t) \\ -\omega \sin \omega t & 0 \end{pmatrix}$$

$$= \begin{pmatrix} \cos(\omega t) & \omega^{-1} \sin(\omega t) \\ -\omega \sin(\omega t) & \cos(\omega t) \end{pmatrix}$$

Solución

$$X(t) = e^{At} X_0$$

Valor inicial

$$X_0 = (x_0 \quad y_0)^t$$

$$e^{At} = I \cos(\omega t) + A\omega^{-1} \sin(\omega t)$$

$$= \begin{pmatrix} \cos(\omega t) & 0 \\ 0 & \cos(\omega t) \end{pmatrix} + \begin{pmatrix} 0 & \omega^{-1} \sin(\omega t) \\ -\omega \sin \omega t & 0 \end{pmatrix}$$

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Solución

$$X(t) = e^{At} X_0$$

Valor inicial

$$X_0 = (x_0 \quad y_0)^t$$

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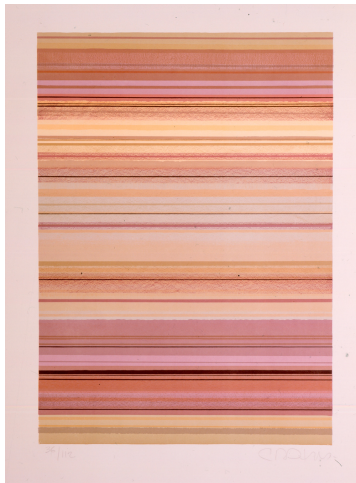
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Por lo tanto

$$\begin{aligned} X(t) &= \begin{pmatrix} \cos(\omega t) & \omega^{-1} \sin(\omega t) \\ -\omega \sin(\omega t) & \cos(\omega t) \end{pmatrix} X_0 \\ &= \begin{pmatrix} x_0 \cos(\omega t) + y_0 \omega^{-1} \sin(\omega t) \\ -\omega x_0 \sin(\omega t) + y_0 \cos(\omega t) \end{pmatrix} \end{aligned}$$

III PARTE: ANÁLISIS COMPLEJO Y ÁLGEBRA LINEAL



Llegó la fecha. Carlos Rojas González.

Demostración del Teorema de Cayley-Hamilton usando el Teorema de Cauchy

Tenemos

- $A \in \mathcal{M}_{n \times n}$.
- $\chi(z) = \det(zI - A) = \sum_{n=0}^N a_n z^n$.

Queremos demostrar

- $\chi(A) = \sum_{n=0}^N a_n A^n = 0$.

Tomemos

$R > 0$ tal que $\|A\| < R$.

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Por un lado

$$\begin{aligned}\frac{1}{2\pi i} \int_{|z|=R} \chi(z)(zI - A)^{-1} dz &= \frac{1}{2\pi i} \int_{|z|=R} \frac{\chi(z)}{z} \left(I - \frac{A}{z}\right)^{-1} dz \\&= \frac{1}{2\pi i} \int_{|z|=R} \chi(z) \left(\sum_{n=0}^{\infty} \frac{A^n}{z^{n+1}}\right) dz \\&= \frac{1}{2\pi i} \sum_{n=0}^{\infty} A^n \int_{|z|=R} \frac{\chi(z)}{z^{n+1}} dz \\&= \frac{1}{2\pi i} \sum_{n=0}^{\infty} A^n \sum_{j=0}^N 2\pi i a_j \delta_{jn} \\&= \sum_{j=0}^N a_j A^j \\&= \chi(A)\end{aligned}$$

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$$\chi(A) = \frac{1}{2\pi i} \int_{|z|=R} \chi(z)(zI - A)^{-1}$$

Por otro lado

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CONCLUSIONES



Reptiles. M. C. Escher.

CONCLUSIONES



Mano con esfera reflectiva. M. C. Escher.

*Muchas
gracias!*

